



TNE10006/TNE60006: Networks and Switching



Transport Layer

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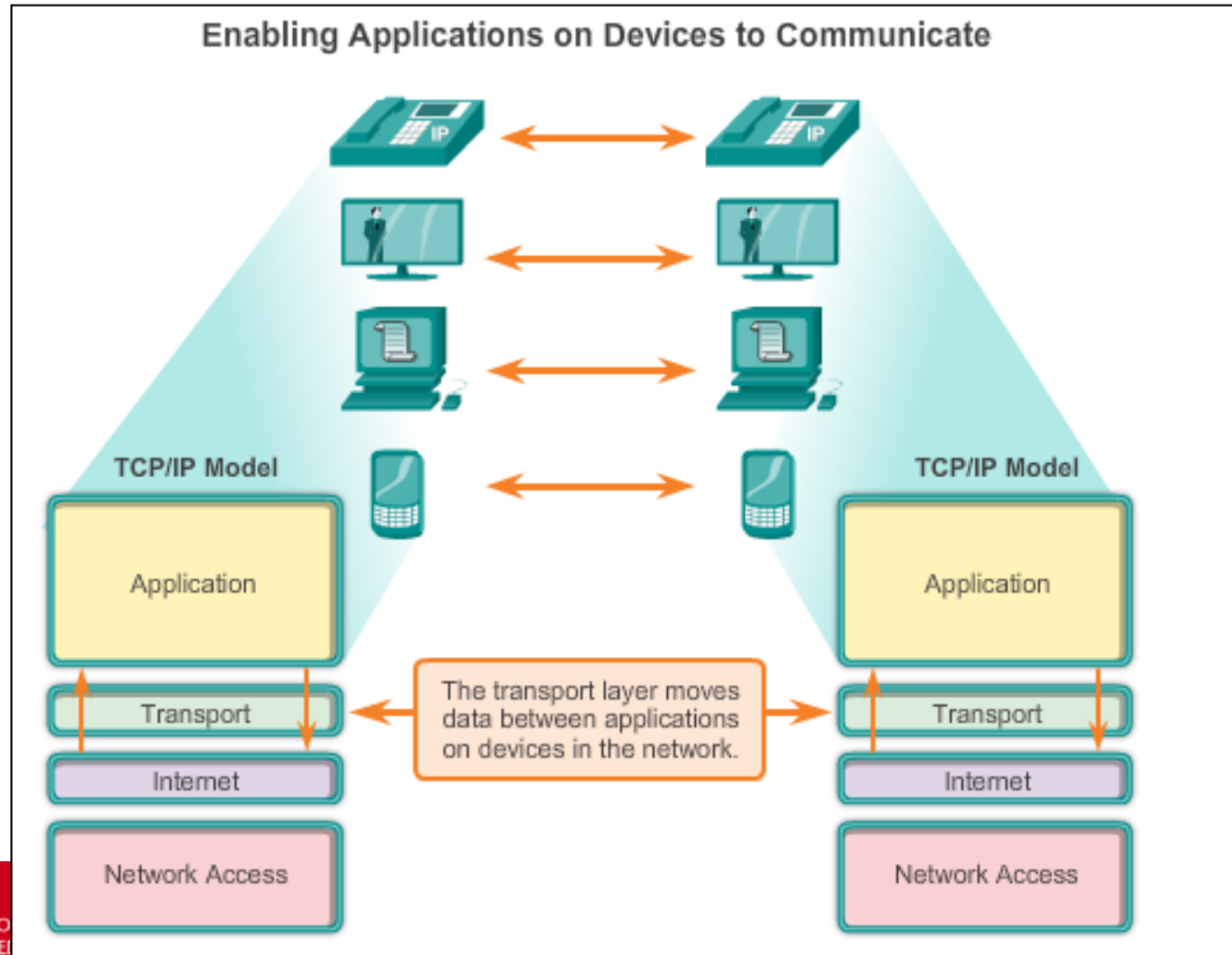
Outline

- Role of the Transport Layer
- Characteristics
- Supporting Concurrent Applications
- Port Numbers



Transport Layer

The Role of the Transport Layer

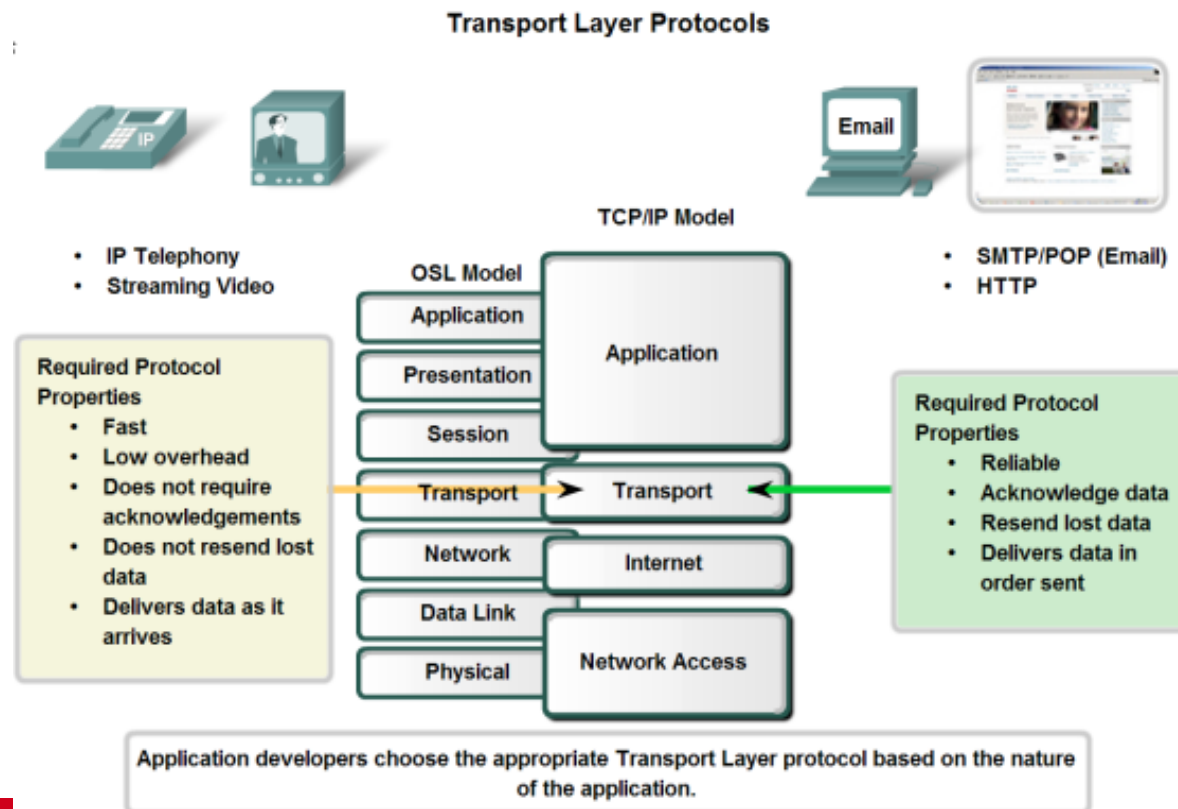




Transport Layer

Basic Characteristics

- Primary role – Allow applications to communicate
- Different categories of applications





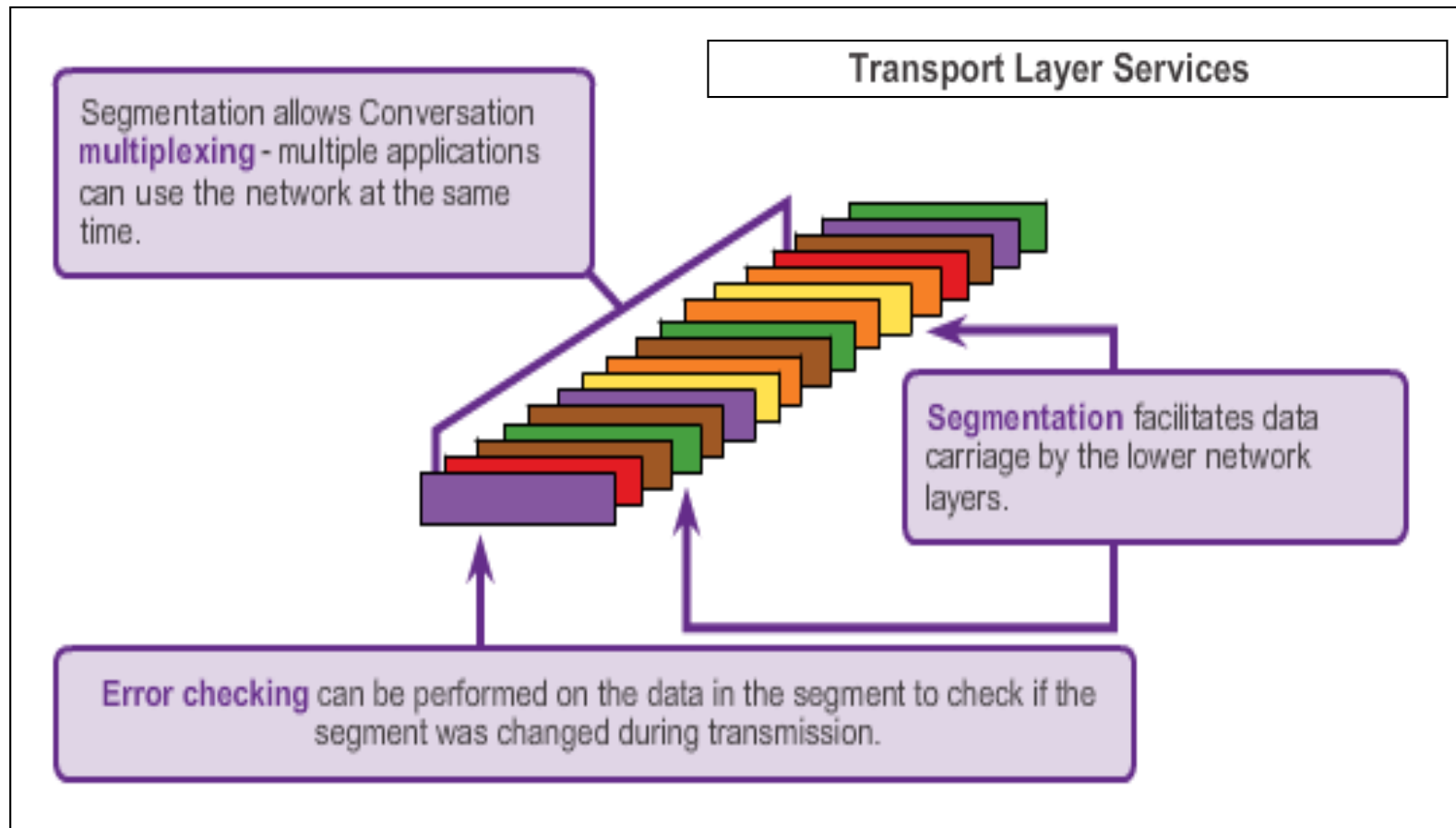
Transport Layer

Basic Characteristics

- Application categories serviced by TCP or UDP
 - UDP – Real-time applications
 - TCP – Reliability required
- Has an impact on
 - How these protocols behave under varying network conditions
 - Subsequent impact on applications
- Other Transport Layer Protocols exist



Transport Layer Supporting Concurrent Applications



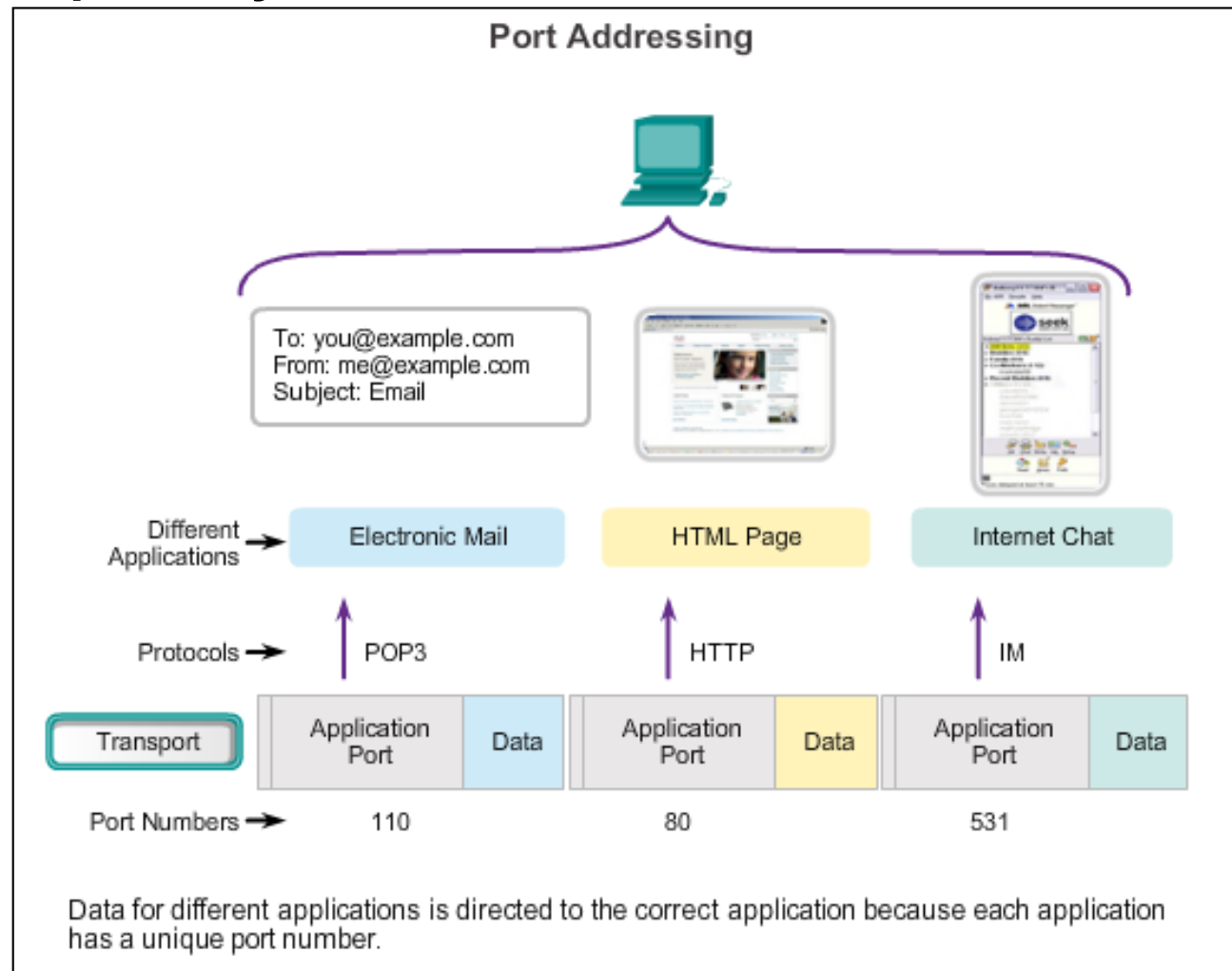


Transport Layer

Supporting Concurrent Applications

- Port Numbers

The Transport Layer Address





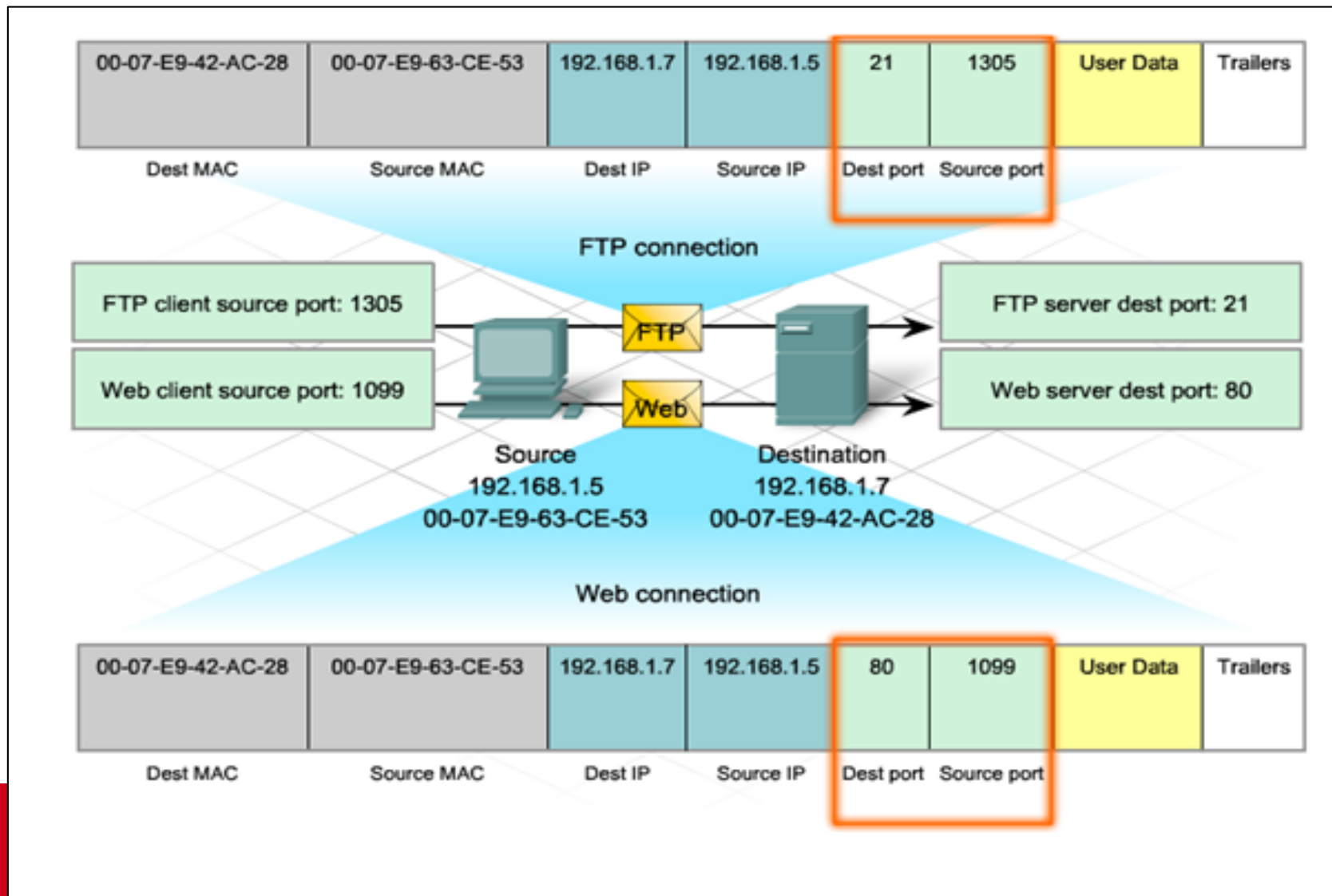
Transport Layer

Supporting Concurrent Applications

- Define a tuple (source/destination IP-address/port-number pairs)
- Uniquely identify separate flows
- For server-type applications
 - Server-side port standardised
 - Client-side port random



Transport Layer Port Numbers





Transport Layer Port Numbers

Port Numbers

Port Number Range	Port Group
0 to 1023	Well Known (Contact) Ports
1024 to 49151	Registered Ports
49152 to 65533	Private and/or Dynamic Ports

Registered TCP Ports:

1863 MSN Messenger
2000 Cisco SCCP (VoIP)
8008 Alternate HTTP
8080 Alternate HTTP

Well Known TCP Ports:

21 FTP
23 Telnet
25 SMTP
80 HTTP
110 POP3
194 Internet Relay Chat (IRC)
443 Secure HTTP (HTTPS)



Transport Layer Port Numbers

Registered UDP Ports:

1812	RADIUS Authentication Protocol
5004	RTP (Voice and Video Transport Protocol)
5040	SIP (VoIP)

Well Known UDP Ports:

69	TFTP
520	RIP

Registered TCP/UDP Common Ports:

1433	MS SQL
2948	WAP (MMS)

Well Known TCP/UDP Common Ports:

53	DNS
161	SNMP
531	AOL Instant Messenger, IRC



Transport Layer Summary

In this lecture, we covered:

- Role of the Transport Layer
- Characteristics
- Supporting Concurrent Applications
- Port Numbers